CSCE 190: Computing in the Modern World

Class 3 – *Sharing Resources via GitHub*

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Carolinian Creed: “I will practice personal and academic integrity.”

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Outline

- Introduction Section
  - Recap from Last Class
- Main Section
  - Sharing Resources – Importance and What to Watch Out For
  - GitHub
  - GitHub pages
- Conclusion Section
  - Ask Me Anything

Introduction Section

Recap from Last Lecture - Class 2

We talked about

- Understanding an Organization
- Understanding Jobs and Career Success
  - Research Career
  - Money v/s Success
- From Course to Careers
- Course Advising

Activity Given – Writing Mission Statement

- Objective: identifying your **personal** mission for a career and success
- Guideline:
  - Think of short (1-3 years), medium (4-7 years) and long-term (8 -) goals
    - While career is the focus, feel free to include personal goals as well
    - Define success criteria for all goals
  - Try to structure how short term (e.g., undergrad program) may help meet **medium-term goals**
  - Try to structure how short and medium may help meet **long-term goals**
- Try to identify organizations and job roles of interest
- Strategize how your major and courses may help

Main Section

Sharing of Resources - Importance

- Growing the ecosystem
  - Collaboration: doing more and faster together
  - Learning: see and emulate from others
- Raising one's own professional stature
  - More the users, more the value of resource (asset) provided

Sharing of Resources – What to Watch Out For

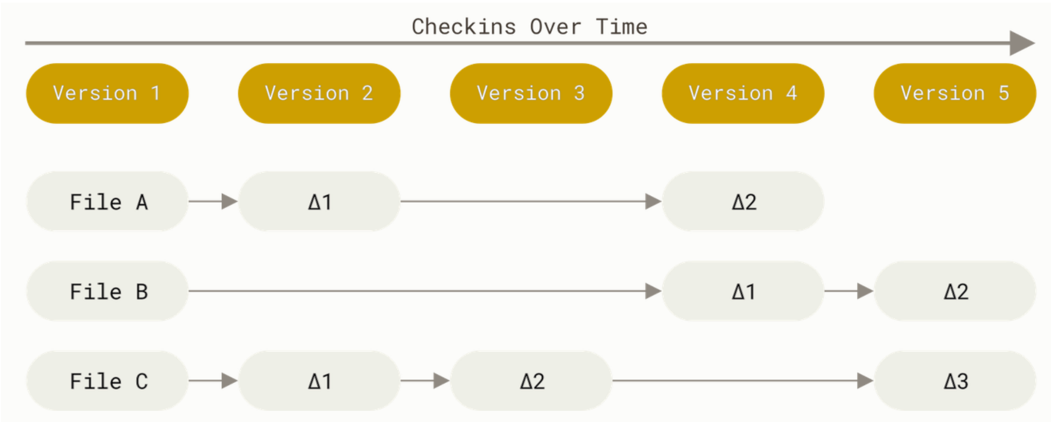
- Intellectual property (IP)
- Privacy
- Unintended commitments

History – Git, GitHub and Significance

- Git
  - Perform version control for files using a (remote, local) server
  - Created by Lunus Torvalds (creator of Linux), 2005 (Source: https://en.wikipedia.org/wiki/Git)
- GitHub
  - Running Git, remote server maintained by a organization (GitHub). Created in 2008.
  - Originally open-source and non-profit, sold in 2018 to Microsoft
- Significance
  - GitHub is the world's largest source code host as of June 2023. Over five billion developer contributions were made to more than 500 million open source projects in 2024.(Source: https://en.wikipedia.org/wiki/GitHub)
  - Works with **any type of files**
- We will use Git and GitHub terms interchangeably for convenience

Git Concepts

- File – anything worth storing
  - Repository – a place with files and folders
- Versions – a file state at a given time
- Commit - taking a snapshot of a repo; push, pull, status
- Branch

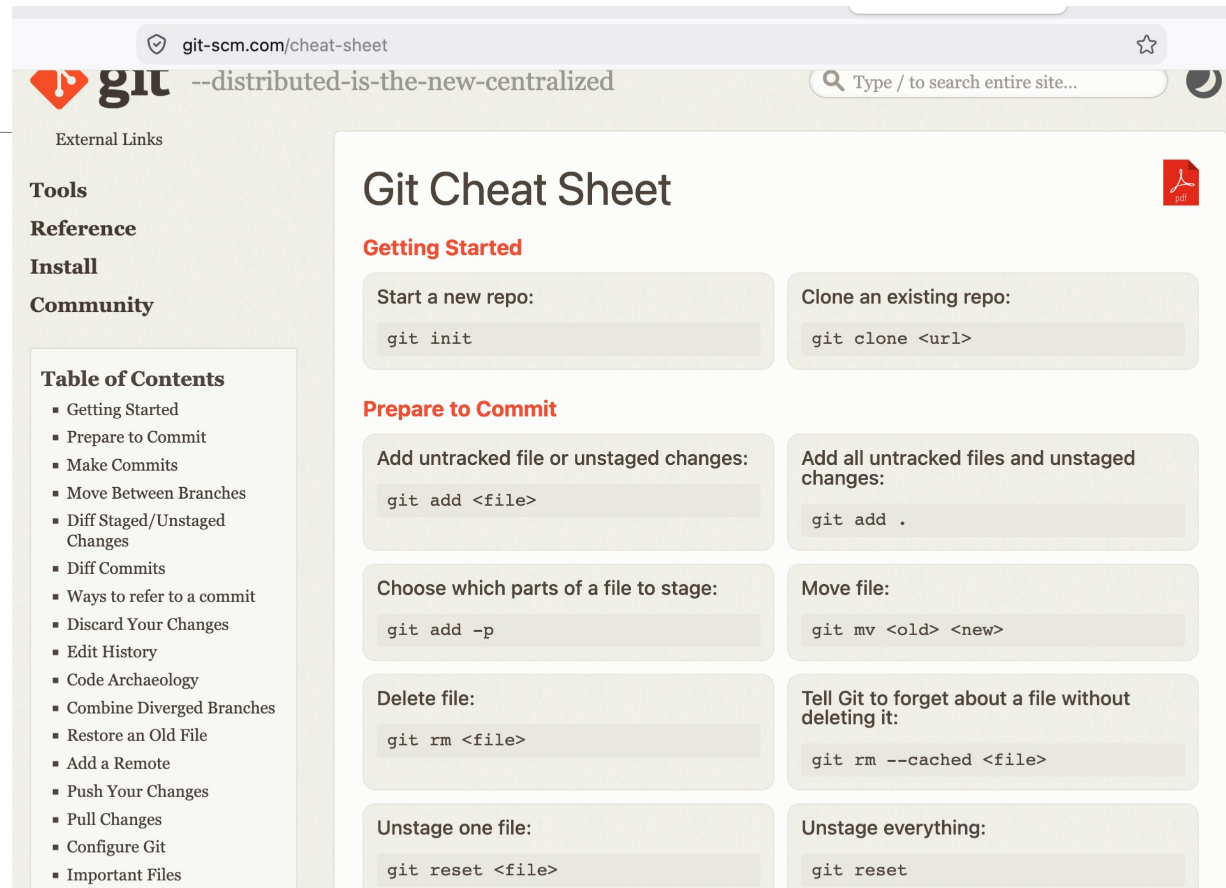


Free book for further reading:

- **Pro Git**, Scott Chacon and Ben Straub (2014), https://git-scm.com/book/en/v2
- Note license: Creative Commons Attribution Non Commercial Share Alike 3.0 license

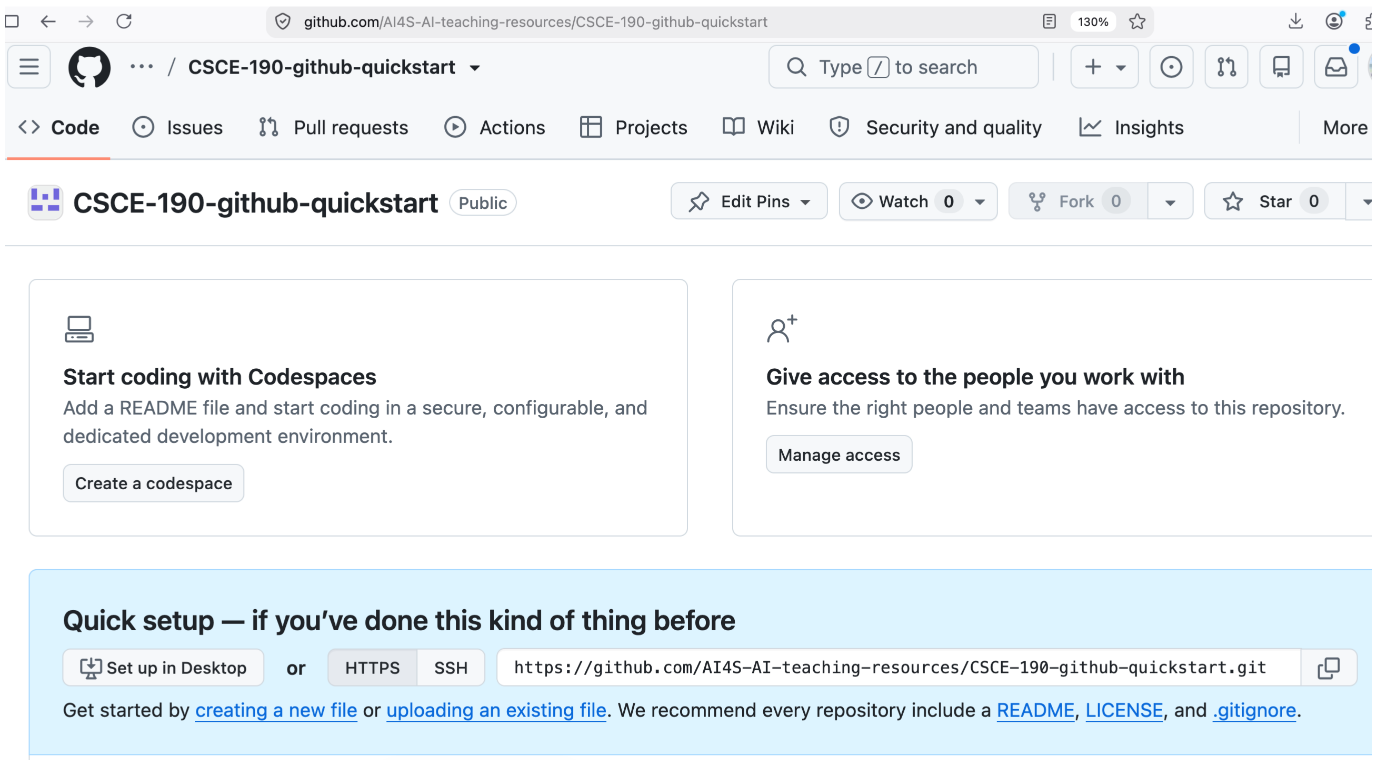
Git Resources

- Cheat sheet: https://git-scm.com/cheat-sheet
- Videos: https://git-scm.com/videos



GitHub Concepts

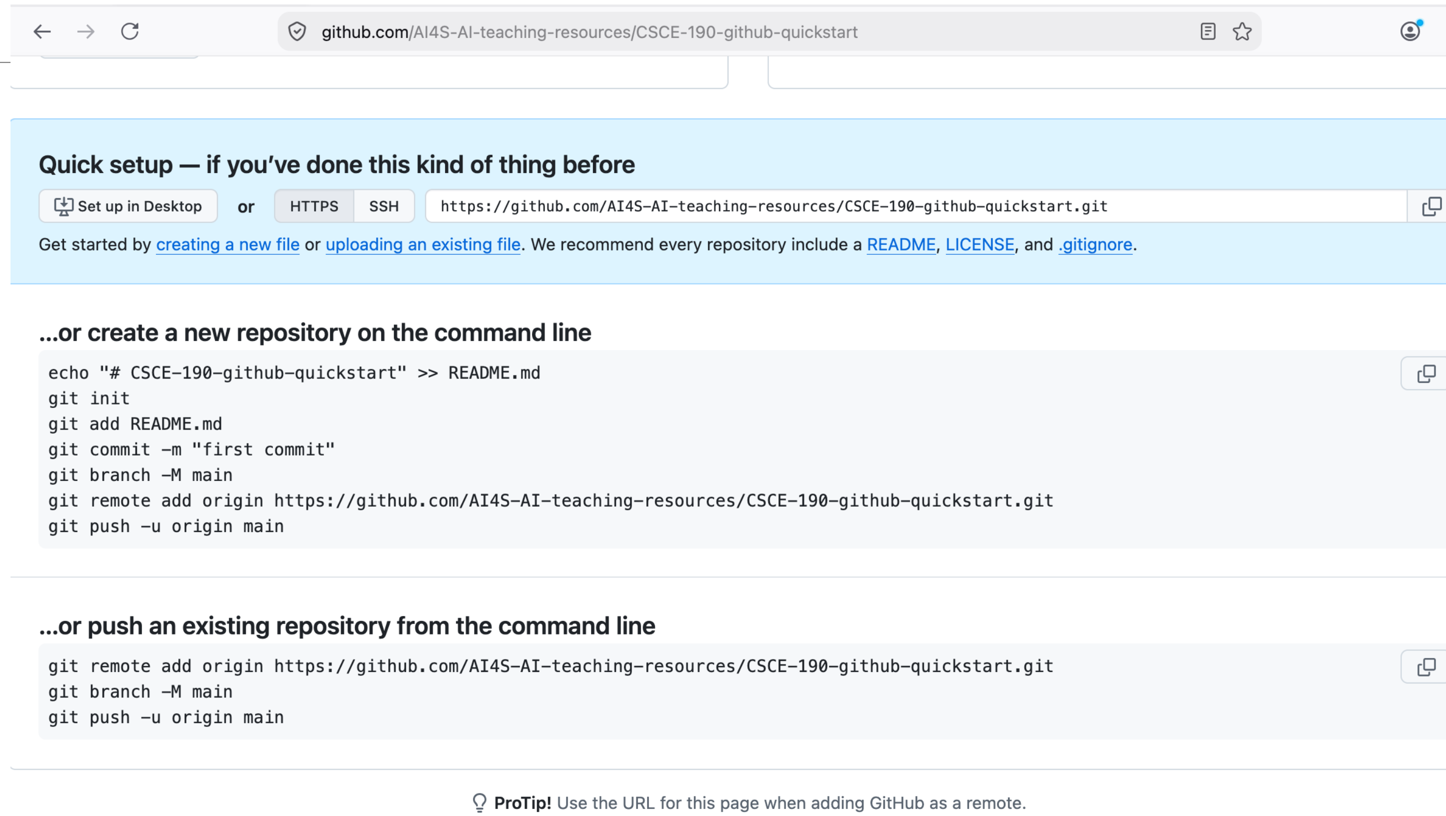
- Organizations
- Projects
- Team



GitHub Concepts

Ways to operate

- Command-line
- GitHub tool
- Using browser

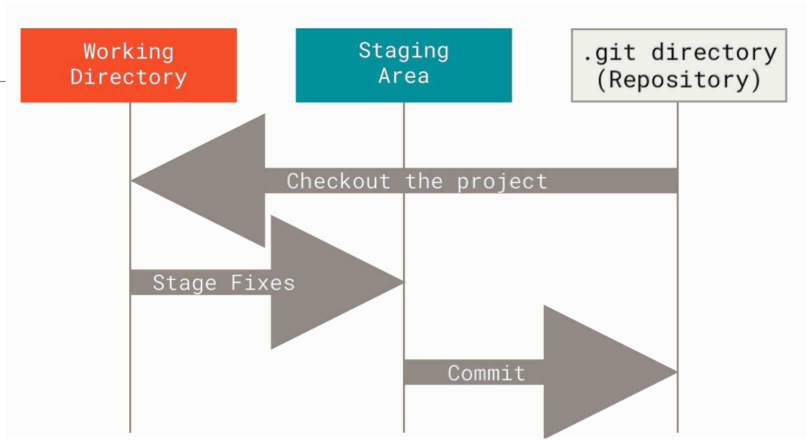


GitHub Usages

- Sharing code (GitHub)
  - Variants: Sharing code and taking actions – e.g., building code-bases periodically
- Creating and hosting web pages (GitHub pages)

GitHub – Managing Code

- Process for making change
  - Make local changes
  - **Add** – add changes to (local) repo
  - **Commit** – tell git that you intend to make changes final
  - **Push** – tell git to make changes to remote server



Source: **Pro Git**, Scott Chacon and Ben Straub (2014), https://git-scm.com/book/en/v2

Demo – Git, GitHub

- Creating an organization
- Creating a repo
  - Adding license
  - Adding .gitignore
  - Cloning a repo
- Creating file
  - Committing changes
  - Pulling a file

GitHub Pages

GitHub Pages

- Using GitHub to host website
  - If you have an account on GitHub, you can create the associate website
  - Uses Git and well-defined file layout to fetch and manage website pages
- Uses HTML and css
- For more, see external presentations (published with permission)
  - GitHub pages – TyB presentation
  - GitHub pages / HTML – RobertB presentation

Document using Markdown

- Background
  - A simple markup language for the web
- Resources
  - Basics: https://www.markdownguide.org/basic-syntax/
  - Cheat sheet: https://www.markdownguide.org/cheat-sheet/

Concluding Section

Conclusion

We talked about

- Sharing Resources – Importance and What to Watch Out For
- GitHub
- GitHub pages
- Touched upon HTML and Markdown

Week	**Content**	**Assignment**
1	Intro, syllabus, university resources	Slack account
2	CSE majors, career paths, research areas	Write a mission statement
3	Github (account and basics)	Github account, repo
4	VS Code, HTML basics	None
5	More HTML, CSS	Update Github with CSS
6	Brainstorming, problem statement	Problem statement
7	Computing – (Software) Dev. Life Cycle	Draft Requirement, Specification
8	Computing – Testing a System	Test Plan
9	Building a resume, LinkedIn	Draft resume in PDF
10	Review resumes	Review resumes, update Github
11	Convert resume to HTML	Begin conversion, update Github
12	Github code repos	Add repos to portfolio, add SDLC outcomes
13	CSE research areas	(Finalize website)
N/A	Last day of classes: 4 Dec, 2026	**All extensions end**

Ask Me Anything